

IMAGE PROCESSING PROJECT

Team: 3 – 5 members

1. What You Will Build

A simple desktop application that lets a user load any image and apply basic image-processing operations through buttons and sliders — no command line needed. You will use Python with OpenCV (image operations) and Tkinter (the window / GUI).

2. Required Features

#	Feature	What it does
1	Open Image	Browse and load a JPEG or PNG file from disk.
2	Show Channels	Display the Red, Green, and Blue channels of the image separately.
3	Grayscale	Convert the image to black-and-white.
4	Geometric Transformation	Zooming (Nearest Neighbor vs. Bilinear) and Rotations.
5	Flip	Flip the image horizontally or vertically.
6	Enhancement	Histogram Equalization and Gamma Correction.
7	Gaussian Blur	Smooth the image. Kernel size controlled by a slider.
8	Edge Detection	Highlight edges using the Canny algorithm.
9	Compare	Display the Output Image and New Histogram simultaneously.
10	Reset	Return to the original loaded image at any time.
11	Save Image	Save the current result as a new file.

3. Useful OpenCV One-Liners

These are the exact functions you need — look them up in the OpenCV docs if you want to understand the parameters:

Operation	Code
Read & convert	<pre>img = cv2.imread('file.jpg') rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)</pre>
Split channels	<pre>b, g, r = cv2.split(img)</pre>
Grayscale	<pre>gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)</pre>
Rotate 90°	<pre>rot = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)</pre>
Flip horizontal	<pre>flipped = cv2.flip(img, 1)</pre>
Gaussian Blur	<pre>blur = cv2.GaussianBlur(img, (11,11), 0)</pre>

Canny edges	<code>edges = cv2.Canny(gray, 100, 200)</code>
-------------	--

4. Setup & Installation

Python 3 must be installed. Run the following command once to install the required libraries: `pip install opencv-python pillow`

Tkinter comes built-in with Python. Pillow is needed to display OpenCV images inside a Tkinter window.

5. Deliverables

- One Python file (or a small folder) — commented so anyone can read it.
- Report explaining the Code.
- A live demo: run the app in front of the instructor and show all features working.
- Each team member should be able to explain at least one feature.

6. Grading

Criterion	Weight
All 9 features work correctly	50%
GUI is clean and easy to use	20%
Code is readable and Report Printed	15%
Live demo & explanation	15%

Keep it simple — a working app beats a half-finished complex one.